KARTHIK RAJ S A

Senior Web3 Engineer • Blockchain Developer • Full-Stack Web2 Architect

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Bangalore, Karnataka, India Open to Remote Worldwide

PROFESSIONAL SUMMARY

Senior Full-Stack Developer with **5+ years** of experience architecting and shipping production-grade **Web3, Blockchain, and Web2 applications**. Deep expertise across the full decentralized stack — Solidity/Rust smart contract development, DeFi protocol engineering, NFT ecosystems, Layer-2 scaling solutions, and cross-chain bridge architecture — combined with robust Web2 foundations in AWS cloud infrastructure, microservices, and API design. Security-first mindset backed by **OSCP/OSCP+** credentials ensures every smart contract and system ships audited and hardened. Proven track record of leading complex protocol integrations, guiding engineering teams toward secure on-chain/off-chain architectures, and delivering products used by millions of users.

PROFESSIONAL EXPERIENCE

Senior Blockchain Engineer (Lead) — American Express

Remote, United States

April 2025 – Present

Network Intelligence Pvt Ltd

- Selected as one of only **five engineers in 26 years** at Network Intelligence to lead Web3 integration at American Express, architecting blockchain-based financial settlement and tokenization infrastructure.
- Designed and deployed **ERC-20 and ERC-1155 smart contracts** for rewards tokenization on a permissioned Ethereum network, reducing settlement from T+2 to near real-time.
- Architected a **cross-chain bridge** (Ethereum mainnet ↔ Optimism L2 rollup) with cryptographic finality guarantees and automated liquidity rebalancing.
- Built **Node.js/TypeScript services** (Ethers.js, Web3.js) for transaction lifecycle management, gas optimization, wallet custody, and on-chain analytics via The Graph.
- Led **smart contract security audits** using Slither and Echidna — remediating reentrancy, overflow, and access-control vulnerabilities before mainnet deployment.

Web3 Subject Matter Expert & Protocol Engineer

Remote, United Kingdom

July 2024 – April 2025

Hack The Box

- Led design and delivery of **Web3 CTF challenge infrastructure** for **3.5M+ researchers** globally — smart contract exploit labs covering flash loans, MEV, and price oracle manipulation.
- Built **automated vuln-detection pipelines** (Slither + Semgrep) integrated into CI/CD, improving detection accuracy by **40%** and significantly reducing false positives.
- Engineered **Foundry/Hardhat/Anvil EVM testnet environments** for DeFi simulation; built IPFS-backed NFT minting pipelines and authored Solidity/EVM internals documentation.

Full-Stack Web3 Developer

Remote, United Kingdom

May 2022 – July 2024

Hack The Box

- Designed and deployed **NFT marketplace contracts** (ERC-721/ERC-1155, EIP-2981, Merkle whitelist) processing over **$2M in on-chain transactions** with zero exploits post-launch.
- Implemented a **DeFi yield aggregator** integrating Uniswap V3, Aave V3, and Compound III with custom Solidity vaults, rebalancing strategies, and automated liquidity hooks.
- Built **React + Wagmi + RainbowKit** dApps with multi-wallet support and ENS resolution; developed The Graph subgraphs cutting data-fetch latency by **80%**.
- Integrated **Polygon, Arbitrum, and Optimism L2 solutions** into Web2 backends, reducing end-user gas costs by **95%** while preserving Ethereum mainnet security.
- Wrote **Chainlink VRF/CCIP oracle integrations** and architected AWS infrastructure (Lambda, API GW, DynamoDB, S3) for off-chain metadata and event indexing.

Independent Web3 Researcher & Bug Bounty Hunter

Remote

October 2020 – March 2025

HackerOne, BugCrowd, Immunefi, Code4rena

- Achieved **Top 20 ranking in India on HackerOne** by identifying critical smart contract and Web3 vulnerabilities across Fortune 500 and DeFi protocol programs.
- Earned **Hall of Fame at NASA (BugCrowd)** and Immunefi recognition for discovering high-impact vulnerabilities in blockchain-adjacent infrastructure and API layers.
- Submitted **high/medium severity findings on Code4rena and Sherlock** — including a flash loan AMM exploit that **protected $8M in user funds**.
- Member of **Team bi0s** — India's #1 CTF team and Top 5 worldwide (CTFtime), specializing in EVM opcode challenges and Solidity bytecode reversing.

TECHNICAL SKILLS

Blockchain & Web3: Solidity, Rust (Anchor/Solana), Vyper, EVM Internals, Cairo (StarkNet)

Smart Contract Tools: Hardhat, Foundry, Truffle, OpenZeppelin, Slither, Echidna, Mythril, Certora

DeFi & Protocols: Uniswap V2/V3, Aave V3, Compound, Curve, Chainlink VRF/CCIP, EIP-4337, ERC-4626

NFT & Token Standards: ERC-20, ERC-721, ERC-1155, EIP-2981, EIP-712, Merkle Proofs, Lazy Minting

Layer 2 & Scaling: Polygon, Arbitrum, Optimism, zkSync, Base, Scroll, State Channels

Indexing & Data: The Graph (AssemblyScript), Moralis, Alchemy, Infura, QuickNode, Dune Analytics

Frontend (dApps): React, Next.js, TypeScript, Wagmi, Ethers.js, Web3.js, RainbowKit, Viem

Backend & APIs: Node.js, Python, Express, FastAPI, GraphQL, REST, WebSockets, gRPC

Cloud & DevOps: AWS (Lambda, EC2, S3, DynamoDB, API GW), Docker, Kubernetes, GitHub Actions, Terraform

Storage & Identity: IPFS, Filecoin, Arweave, ENS, Ceramic, Lit Protocol, WalletConnect

Databases: PostgreSQL, MongoDB, Redis, DynamoDB, Supabase

Security: Smart Contract Auditing, Reentrancy, Flash Loan Vectors, Formal Verification, SAST/DAST

CERTIFICATIONS

OFFENSIVE SEC
OSCP
Offensive Security

OFFENSIVE SEC
OSCP+
Offensive Security

CLOUD SEC
AZ-500
Azure Security Eng.

CLOUD SEC
CCSP
(ISC)^2 Training

APP SEC
CPTS I & II
Hack The Box

RED TEAM
CRTO I, II & III
Zero-Point Security

NETWORK SEC
CNPen
Network Intelligence

NETWORK SEC
Fortinet NSE III
Fortinet

MOBILE & API
Mobile & API Penetration Testing
TCM Security

COMPLIANCE
ISO/IEC 27001 Info Sec Associate
PECB

KEY ACHIEVEMENTS & IMPACT

- **Top 5 India, Top 100 Globally** on HackTheBox; solved *AWS Fortress* — 1 of only 150 successful solvers out of 3.5M hackers worldwide.
- Deployed **NFT and DeFi contracts processing $2M+ in on-chain transactions** across Ethereum mainnet and L2 networks with zero post-deployment exploits.
- Achieved **60% gas cost reduction** across three production DeFi protocols through EVM opcode optimization, storage packing, and calldata compression.
- Immunefi flash loan vulnerability finding **protected $8M in user funds**; high/medium findings on Code4rena and Sherlock audit contests.
- Built and open-sourced a **Foundry ERC-4626 invariant testing framework** adopted by 200+ developers in Ethereum developer communities.
- **Team bi0s** member — India's #1 CTF team, Top 5 worldwide (CTFtime); NASA Hall of Fame on BugCrowd.

EDUCATION

2021

Bachelor of Engineering in Computer Engineering

VTU University, Bangalore, India

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